AI & ML Lab Assignment Report

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Course Code: STA 0542 5184

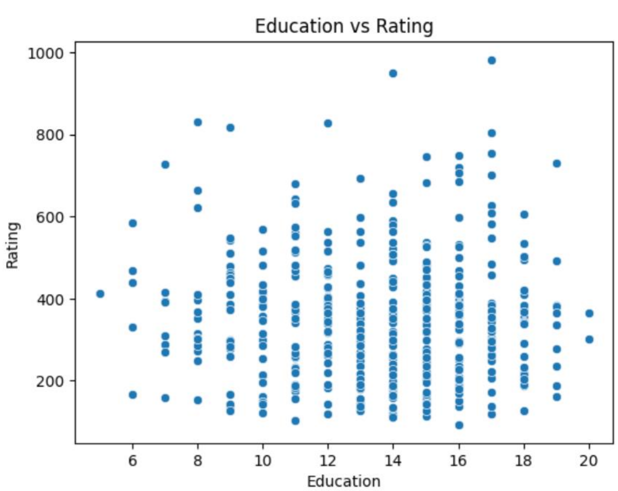
Question 01 — Simple Linear Regression (Credit Data)

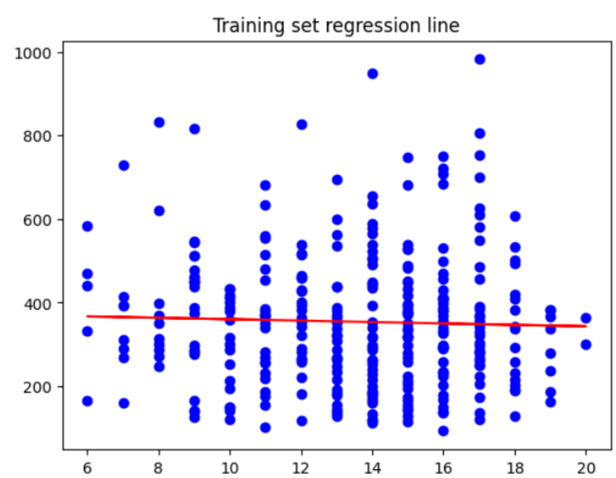
Task: Predict *Rating* using *Education*.

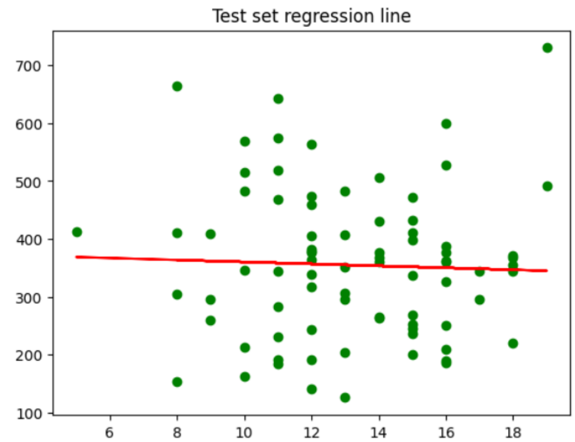
Results

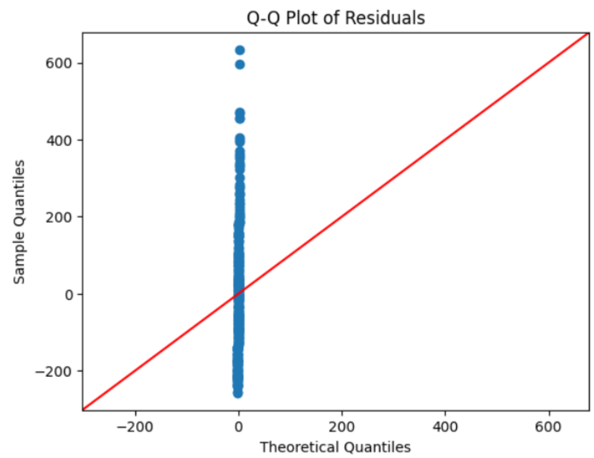
- OLS Regression Results:
  - Intercept = 377.28
  - Coefficient (Education) = –1.69
  - p-value = 0.552 (not significant)
  - $R^2 = 0.001$, Adjusted $R^2 = -0.002$
- Evaluation Metrics:
  - MSE = 16,390.61
  - RMSE = 128.03
  - MAE = 100.20
- Residual diagnostics: Q-Q plot shows deviations from normality.

Interpretation: Education does not significantly predict Rating. The coefficient is negative but not statistically significant. The model explains almost none of the variance ($R^2 \approx 0.001$), and prediction errors are large.









Question 02 — Classification (Credit Data)

Task: Classify *Own* using Rating, Student, Age, Married, Limit.

Results

Confusion Matrices:

- Logistic Regression: [[10, 26], [20, 24]]
- Decision Tree: [[16, 20], [18, 26]]
- Random Forest: [[22, 14], [19, 25]]
- KNN: [[16, 20], [20, 24]]
- SVM: [[20, 16], [29, 15]]

Metrics comparison:

Model	Accuracy	Precision	Recall	F1	MAE	RMSE
Logistic Regression	0.425	0.480	0.546	0.511	0.575	0.758
Decision Tree	0.525	0.565	0.591	0.578	0.475	0.689
Random Forest	0.588	0.641	0.568	0.602	0.413	0.642
KNN	0.500	0.545	0.545	0.545	0.500	0.707
SVM	0.438	0.484	0.341	0.400	0.563	0.750

Interpretation: Random Forest achieved the highest accuracy (≈ 0.59) and balanced precision/recall, making it the best classifier among the tested models. Overall accuracies are modest, indicating the classification task is challenging with the given features.

Question 03 — Regression (Credit Data)

Task: Predict *Rating* using Income, Limit, Age, Own, Student, Married.

Results

Metrics comparison:

Model	MSE	RMSE	MAE	R^2
Decision Tree	258.26	16.07	13.36	0.9842
Random Forest	200.91	14.17	11.45	0.9877
KNN	3737.40	61.13	39.79	0.7718

SVM	15195.0	123.27	96.69	0.0721
GradientBoost	190.43	13.80	11.22	0.9884
AdaBoost	192.63	13.88	11.32	0.9882
XGBoost	202.17	14.22	11.03	0.9877

Lasso Regression:

- Alpha = 0.168
- MSE = 122.41, $R^2 = 0.993$
- Selected features: Income, Limit, Married

Ridge Regression:

- Alpha = 0.1
- MSE = 128.26, $R^2 = 0.992$
- Important features: Income, Limit, Married

Interpretation: GradientBoost achieved the best performance ($R^2 \approx 0.988$, lowest MSE). Random Forest and XGBoost also performed excellently. Lasso and Ridge confirm that Income, Limit, and Married status are the most influential predictors of Rating.

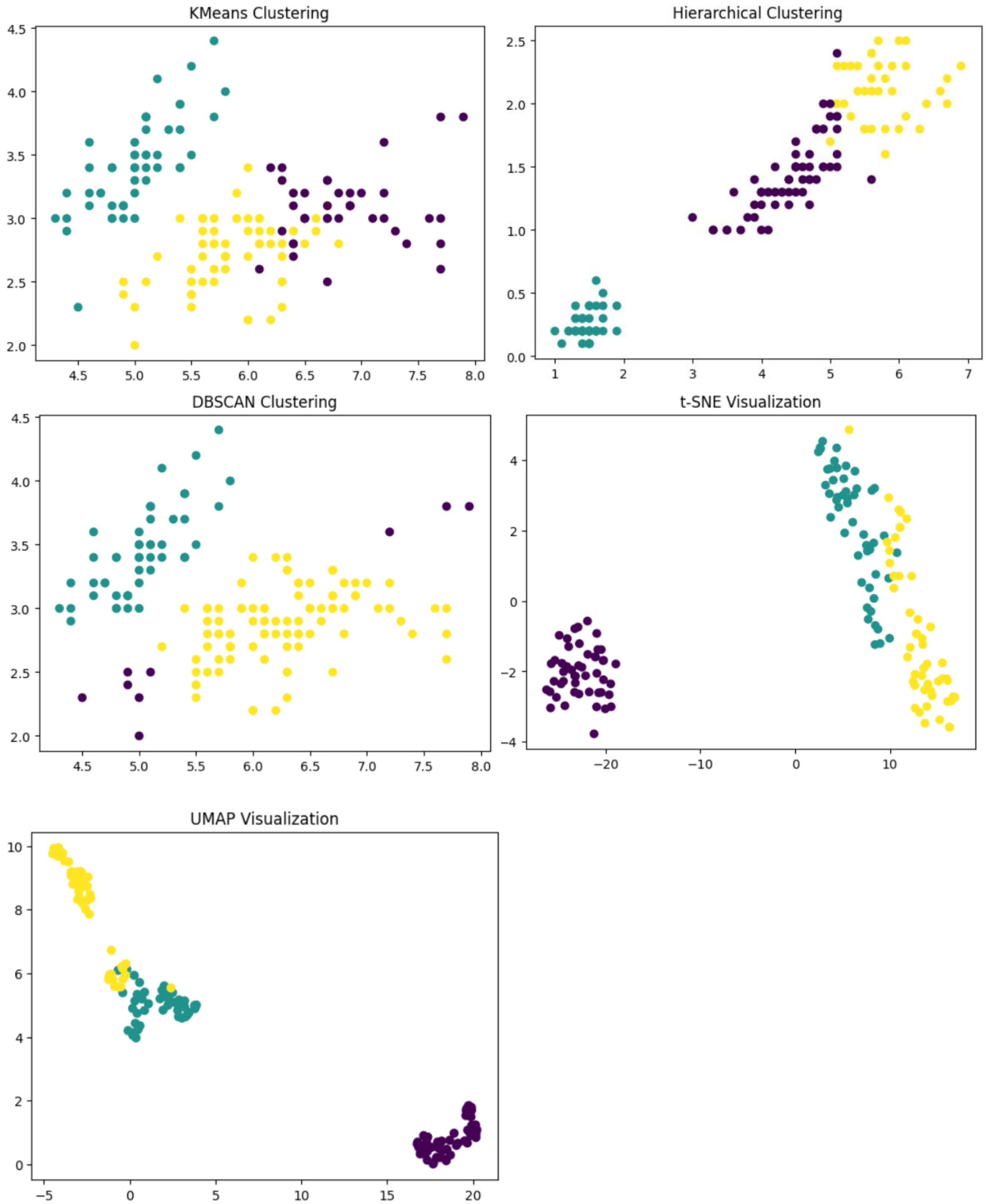
Question 04 — Clustering & Dimensionality Reduction (Iris Data)

Task: Cluster

Results

- **KMeans Clustering:** Clear separation of Setosa, overlap between Versicolor and Virginica.
- **Hierarchical Clustering:** Ward linkage separates Setosa cleanly, Versicolor and Virginica overlap.
- **DBSCAN:** Identifies clusters with some noise points; Setosa cluster is distinct.
- **t-SNE Visualization:** Shows clear separation of Setosa, partial overlap of Versicolor and Virginica.
- **UMAP Visualization:** Similar to t-SNE, with slightly better separation of Versicolor and Virginica.

Interpretation: Clustering methods consistently separate Setosa well, while Versicolor and Virginica overlap. Dimensionality reduction (t-SNE, UMAP) improves visualization of species clusters.


KMeans Clustering
4.5
4.0
3.5
3.0
2.5
2.0
4.5 5.0 5.5 6.0 6.5 7.0 7.5 8.0
Hierarchical Clustering
2.5
2.0
1.5
1.0
0.5
0.0
1 2 3 4 5 6 7
DBSCAN Clustering
4.5
4.0
3.5
3.0
2.5
2.0
4.5 5.0 5.5 6.0 6.5 7.0 7.5 8.0
t-SNE Visualization
4
2
0
−2
−4
−20 −10 0 10
UMAP Visualization
10
8
6
4
2
0
−5 0 5 10 15 20